Mining pool network and decentralization of a PoW blockchains

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Abstract

The Proof-of-Work (PoW) blockchain relies on miners who incur constant operational costs while receiving stochastic rewards. To mitigate financial uncertainty, many miners join Pay-Per-Share (PPS) mining pools, which provide stable payouts while transferring risk to pool managers. This paper develops a two-stage analysis of miners' strategic behavior and pool competition. In both approaches, the pool manager's surplus is modeled as a two-sided jump process. First, using a mean-variance trade-off approach, we characterize the optimal allocation of miners across mining pools and investigate Nash equilibrium among pools. Second, we study the pool manager's payout problem within an expected discounted dividend framework. Using techniques from actuarial risk theory, we establish and prove the optimality of a barrier strategy under specific conditions and provide numerical methods to compute the optimal barrier and associated dividends. We then analyze the corresponding Nash equilibrium in fees under this criterion. Our findings contribute to understanding how economic incentives influence miner participation and the decentralization of PoW blockchain networks.

1 Introduction

The Proof-of-Work (PoW) blockchain consensus mechanism, as first introduced by Nakamoto [2008], forms the foundation of decentralized cryptocurrencies such as Bitcoin. This system relies on miners who expend computational resources to validate transactions and secure the network. However, mining is characterized by a dual-risk structure: miners face a constant operational cost (electricity, hardware maintenance) while their rewards—block subsidies and transaction fees—are inherently

stochastic and infrequent. This financial uncertainty poses a significant challenge, as it directly impacts the sustainability of individual miners.

To mitigate this volatility, miners often choose to join mining pools. These pools aggregate computational power, distribute rewards more predictably, and ensure more stable revenue streams. Among the various pooling strategies, the Pay-per-Share (PPS) model [Albrecher et al., 2022; Rosenfeld, 2011; Schrijvers et al., 2016] provides miners with a fixed payout per valid share they contribute, effectively transferring risk to the pool manager.

Miners must make a strategic decision regarding participation in a mining pool, balancing expected revenue against the risk of ruin. This decision can be analyzed using an expected discounted dividend approach, which evaluates the long-term viability of miners given their financial constraints. In this context, optimal dividend strategies, such as the barrier strategy [Avanzi et al., 2007], emerge as critical tools for sustaining miner operations and maximizing profitability. These strategies rely on results from fluctuation theory and scale functions [Bayraktar et al., 2013] to determine optimal capital retention and payout policies.

While extensive research exists on individual miner strategies, the surplus process of the pool manager remains relatively unexplored. It can be modeled as a two-sided jump process, incorporating incoming miner contributions and outgoing payouts. However, existing literature provides limited results on optimal surplus management in this setting. In this paper, we establish that a band strategy is optimal in general and that a simple barrier strategy becomes under certain restrictions. We further develop numerical methods to identify the optimal barrier level and compute corresponding dividends.

Beyond individual miner and pool manager strategies, we leverage our findings to analyze the equilibrium distribution of miners across available mining pools. By considering a set of miners and a set of pool managers, we derive insights into how miners allocate their computational power in response to different payout schemes and risk profiles. Our study further extends to a numerical investigation of Nash equilibria, illustrating the strategic interactions between miners and pool managers in a competitive setting.

This research contributes to the ongoing discourse on decentralization in PoW blockchains. By providing a framework for mining pool strategies and decision-making, we offer insights into how economic incentives shape miner participation, thereby influencing the overall decentralization and stability of blockchain networks.

2 Wealth processes for miners and pool managers

2.1 miners' wealth process

As introduced by Albrecher and Goffard [2022], The miners' wealth can be modeled as a stochastic process

$$X_t = x - c \cdot t + N_t \cdot b, \text{ for } t \geq 0, \tag{1}$$

where $x > 0$ represents the initial capital, c is the operational cost, b is the reward for discovery of blocks and $(N_t)_{t\geq 0}$ is a Poisson process with intensity λ, which represents the number of blocks found per unit of time. The variance of this process exposes miners to the risk of insolvency. To reduce income volatility, miners join mining pools, aggregating their computational power to increase the frequency of rewards and obtain more stable earnings. A pool manager oversees this collective activity by assessing each miner's contribution and allocating rewards accordingly. This paper considers the Pay-per-Share (PPS) scheme which is a widely used reward mechanism. Under this system, miners submit shares as proof of their work. A share corresponds to a solution of a cryptographic puzzle with a lower difficulty than that required to mine a full block. A PPS deal is characterized by a difficulty reduction $\delta \in (0,1]$ and a pool fee $f \in [0,1)$. In the PPS framework, the pool manager pays a fixed reward $b_k < b$ to miners for each submitted share. Then the stochastic process that represents the miners' wealth becomes

$$\tilde{X}_t = x - c \cdot t + \tilde{N}_t \cdot b_k, \text{ for } t \geq 0, \tag{2}$$

where $b_k = b \cdot \delta \cdot (1-f)$ is the share reward and $(\tilde{N}_t)_{t\geq 0}$ is a Poisson process with intensity $\tilde{\lambda} = \frac{\lambda}{\delta} \geq \lambda$, which exceeds that of $(N_t)_{t\geq 0}$ in (1). The advantage of the PPS system is that it transfers risk from miners to pool managers. Equations (1) and (2) allow Albrecher et al. [2022] to quantify miners' risk through ruin probabilities. Note that solo mining can be interpreted as a particular case of the PPS framework. Indeed, it corresponds to a contract with no fee and no difficulty reduction, that is, $f = 0$ and $\delta = 1$. In this case, the share reward reduces to $b_k = b$ and the arrival intensity becomes $\tilde{\lambda} = \lambda$. Consequently, equation (2) simplifies to

$$\tilde{X}_t = x - c \cdot t + \tilde{N}_t \cdot b, \quad t \geq 0, \tag{3}$$

which coincides with (1). Hence, solo mining can be viewed as a particular case of PPS contract.

2.2 Pool manager's wealth process

Consider $m \in \mathbb{N}$ mining pools, each offering a PPS contract characterized by a difficulty reduction δ_k and a fee f_k, for $k = 1, \ldots, m$. A PPS pool k attracts miners by offering a contract (δ_k, f_k). These contractual parameters determine both the effective share arrival rate and the associated reward. More precisely, for a miner with hashrate λ, participation in pool k leads to a share arrival intensity

λ/δ_k and a reward per share $b_k = b\,\delta_k(1-f_k)$. This contract increases the frequency of rewards while reducing their size, thereby lowering the variance of the miner's wealth at the cost of a lower expected payoff. In addition, the miner retains the possibility to mine solo, which can be interpreted as an artificial pool indexed by $k=0$, with $\delta_0 = 1$, $f_0 = 0$, and thus $b_0 = b$. The pool k collects a total hashpower $H_k = \sum_{i\in\mathcal{I}} \lambda_i$, where $\mathcal{I}$ denotes the set of miners who joined pool k and λ_i is the hashpower contributed by miner i. The rate of arrival of shares of pool k is given by

$$\mu_k = \frac{H_k}{\delta_k},$$

which corresponds to the average number of shares found by the pool participants per time unit. Each share has probability $\delta_k/(1+\delta_k)$ of being an actual solution to the cryptographic puzzle, and probability $1/(1+\delta_k)$ of being a simple share. The wealth process of the pool manager is given by

$$Y_t = y + \sum_{i=1}^{M_t} V_i, \tag{4}$$

where

- $(M_t)_{t\geq 0}$ is a Poisson process with intensity μ_k,
- the V_i's are i.i.d. with probability distribution

$$\mathbb{P}(V=-b_k) = \frac{1}{1+\delta_k}, \qquad \mathbb{P}(V=b-b_k) = \frac{\delta_k}{1+\delta_k},$$

  where $b_k = b(1-f_k)\delta_k$ is the reward that the pool k offers.

Remark 1. *Using the superposition, one can express the surplus of the pool manager (4) as the difference between two Poisson processes associated to deterministic positive and negative jumps.*

Each time a share arrives, the pool manager pays the miner the fixed PPS amount $b(1-f_k)\delta_k$, regardless of whether the share is a solution or not. When the share turns out to be an actual solution, the manager additionally collects the block reward b. The pool is profitable in expectation if and only if

$$\mathbb{E}[V] > 0, \tag{5}$$

which determines admissible combinations of (δ, f). This formulation makes explicit that the PPS contract transfers risk from miners to the pool manager: increasing the frequency of rewards reduces miners' variance, while concentrating risk in the manager's wealth process.

3 Competition between pools with mean-variance preferences

We consider a mining ecosystem composed of a population of heterogeneous miners and two PPS pool managers. The interaction is modeled as a static game in which for a given difficulty reduction $(\delta_1, \delta_2) \in (0,1]^2$, pool managers compete through fees, anticipating the participation decisions of miners who evaluate available options according to a mean–variance criterion.

3.1 Miners' mean-variance

Consider a population of n miners, indexed by $i = 1, \ldots, n$. Each miner i is characterized by a block discovery rate $\lambda_i > 0$, an operational cost $c_i \geq 0$, and a risk-aversion parameter $\gamma_i > 0$. The miner can choose between solo mining, indexed by $k = 0$, and two PPS pools, indexed by $k = 1, 2$. For a given contractual pair (f_k, δ_k), the miner's wealth process over a time horizon T follows the PPS formulation (2) introduced in Section 2.1. The miner evaluates option k according to the mean–variance criterion

$$S_i(k) := \mathbb{E}\left[X_T^{(i,k)}\right] - \gamma_i \operatorname{Var}\left(X_T^{(i,k)}\right). \tag{6}$$

The expected wealth is

$$\mathbb{E}\left[X_T^{(i,k)}\right] = x - c_i T + T \frac{\lambda_i}{\delta_k} b(1 - f_k)\delta_k = x - c_i T + T \lambda_i b(1 - f_k), \tag{7}$$

while the variance is

$$\operatorname{Var}\left(X_T^{(i,k)}\right) = T \frac{\lambda_i}{\delta_k} b^2 (1 - f_k)^2 \delta_k^2 = T \lambda_i b^2 (1 - f_k)^2 \delta_k. \tag{8}$$

Hence,

$$S_i(k) = x - c_i T + T \lambda_i \left(b(1 - f_k) - \gamma_i b^2 (1 - f_k)^2 \delta_k\right). \tag{9}$$

Since the additive term $x - cT$ and the multiplicative term T are common across options, the miner's decision is independent of the horizon T.

For solo mining, we set $(f_0, \delta_0) = (0, 1)$, which gives

$$S_i(0) = \lambda_i \left(b - \gamma_i b^2\right).$$

For participation in pool $k \in \{1, 2\}$,

$$S_i(k) = \lambda_i \left(b(1 - f_k) - \gamma_i b^2 (1 - f_k)^2 \delta_k\right).$$

The miner therefore chooses

$$k_i^* \in \arg\max_{k \in \{0,1,2\}} S_i(k). \tag{10}$$

Because $\lambda_i > 0$ is a common multiplicative factor across all options, it does not affect the ranking of the alternatives. As expressed Albrecher et al. [2025], when we use the mean-variance criterion, miner's decision depends only on γ_i and on the contractual parameters (f_k, δ_k). Thus, miners with identical risk preferences choose the same option and allocate their entire hashrate to the pool that maximizes their objective function.

3.2 Pool managers mean-variance

For a given horizon $T = 1$, the surplus process of pool manager k associated with a contractual pair (f_k, δ_k) is the compound Poisson process (4). Its expectation is given by

$$\mathbb{E}\left[Y_t^{(k)}\right] = y_k + \mu_k \frac{b\delta_k}{1 + \delta_k}\left(f_k(1 + \delta_k) - \delta_k\right), \tag{11}$$

and its variance is given by

$$\mathrm{Var}\left(Y_t^{(k)}\right) = \mu_k \left[\frac{b^2(1-f_k)^2\delta_k^2}{1+\delta_k} + \frac{b^2\delta_k\left(1-(1-f_k)\delta_k\right)^2}{1+\delta_k} \right], \tag{12}$$

where μ_k denotes the share-arrival intensity of pool k, as introduced in Section 2.2. For a fixed difficulty reduction δ_k, pools compete solely through their fee. Consequently, the total hashpower attracted by pool k depends on the fee vector (f_k, f_{-k}), where f_k is the fee set by pool k and $f_{-k} = (f_j)_{j\neq k}$ denotes the fees of competing pools. We therefore write

$$H_k(f_k, f_{-k}) = \sum_{i\in\mathcal{I}_k(f_k,f_{-k})} \lambda_i, \tag{13}$$

where $\mathcal{I}_k(f_k, f_{-k})$ denotes the set of miners who join pool k under the fee profile (f_k, f_{-k}), and λ_i is the hashpower contributed by miner i. Thus the share-arrival intensity of pool k is given by

$$\mu_k = \frac{H_k(f_k, f_{-k})}{\delta_k}. \tag{14}$$

In particular, if all miners join pool k, then $\mathcal{I}_k = \{1,\dots,n\}$ and $H_k(f_k, f_{-k}) = \lambda$, where $\lambda = \sum_{i=1}^n \lambda_i$ denotes the total hashpower available on the network.

The manager evaluates outcomes according to a conditional mean–variance criterion. For a given realization of the aggregate hashrate H_k, the surplus process $Y_t^{(k)}$ is a compound Poisson process with intensity $\mu_k = H_k/\delta_k$. The manager's conditional objective is defined as

$$J_k(H_k) = \mathbb{E}\left[Y_t^{(k)} \mid H_k\right] - \eta_k \,\mathrm{Var}\left(Y_t^{(k)} \mid H_k\right), \tag{15}$$

where $\eta_k > 0$ is the manager's risk-aversion coefficient. Substituting (11), (12) and (14) into (15) yields

$$J_k(H_k(f_k, f_{-k})) = y_k + H_k(f_k, f_{-k})\frac{b}{1+\delta_k}\left[f_k(1+\delta_k) - \delta_k - \eta_k b\left(\delta_k(1-f_k)^2 + \left(1-(1-f_k)\delta_k\right)^2\right)\right]. \tag{16}$$

3.3 Fee competition using expected approach

We now analyze a fee-setting game between the two pool managers. The parameters (δ_1, δ_2) are fixed ex ante and interpreted as technological characteristics of the two pools. Managers choose fees (f_1, f_2), miners then respond by selecting the option that maximizes their mean–variance score (10), and managers receive payoffs given by (16).

3.3.1 Threshold structure of miner choices

We assume that miners are heterogeneous only through their risk-aversion coefficient and that

$$\gamma \sim \mathrm{Unif}[0, \bar{\gamma}], \tag{17}$$

independently across n miners. Since λ_i factors out of all pairwise comparisons, it is convenient to normalize the miner's score by λ_i and define

$$s_0(\gamma) = b - \gamma b^2, \tag{18}$$

$$s_k(\gamma) = b(1 - f_k) - \gamma \delta_k b^2 (1 - f_k)^2, \qquad k \in \{1, 2\}. \tag{19}$$

Each miner selects $k \in \{0, 1, 2\}$ to maximize $s_k(\gamma)$. Since the mean–variance scores are affine in the risk-aversion parameter γ, pairwise differences are affine, implying that optimal choices are characterized by cutoff values of γ. Consequently, the set of types selecting any given option forms an (possibly empty) interval in $[0, \bar{\gamma}]$, and the equilibrium allocation is fully described by a finite collection of such thresholds.

Proposition 1 (Threshold structure)**.** *Consider a miner with risk-aversion parameter $\gamma \in [0, \bar{\gamma}]$. The optimal choice problem (10) is characterized by three thresholds.*

(i) Pool versus solo mining. *For $k \in \{1, 2\}$, pool k is preferred to solo mining if and only if*

$$s_k(\gamma) \geq s_0(\gamma) \quad \Longleftrightarrow \quad \gamma \geq \gamma_{k0}(f_k),$$

where

$$\gamma_{k0}(f_k) = \frac{f_k}{b\,(1 - \delta_k (1 - f_k)^2)}. \tag{20}$$

(ii) Comparison between pools. *Define*

$$D_{12}(f_1, f_2) = \delta_1 (1 - f_1)^2 - \delta_2 (1 - f_2)^2. \tag{21}$$

If $D_{12}(f_1, f_2) \neq 0$, the comparison between the two pools is characterized by the threshold

$$\gamma_{12}(f_1, f_2) = \frac{f_2 - f_1}{b\, D_{12}(f_1, f_2)}, \tag{22}$$

such that

$$s_1(\gamma) \geq s_2(\gamma) \quad \Longleftrightarrow \quad \begin{cases} \gamma \leq \gamma_{12}(f_1, f_2), & \text{if } D_{12}(f_1, f_2) > 0, \\ \gamma \geq \gamma_{12}(f_1, f_2), & \text{if } D_{12}(f_1, f_2) < 0. \end{cases} \tag{23}$$

If $D_{12}(f_1, f_2) = 0$, the comparison between the two pools is independent of γ and depends only on expected returns.

(iii) Partition of the type space. *The interval $[0, \bar{\gamma}]$ is partitioned into at most three subintervals, corresponding respectively to miners who optimally choose solo mining, pool 1, and pool 2.*

Proof. For each $k \in \{0, 1, 2\}$, the score $s_k(\gamma)$ is affine in γ. Hence, for any pair (i, j), the difference $s_i(\gamma) - s_j(\gamma)$ is also affine and admits at most one root. Each pairwise comparison therefore induces a unique threshold value of γ, which establishes the expressions for γ_{k0} and γ_{12}. The partition of the interval $[0, \bar{\gamma}]$ follows directly from these pairwise comparisons. □

Remark 2 (Economic interpretation)**.** *The thresholds $\gamma_{k0}(f_k)$ represent entry conditions into the pooling market: only sufficiently risk-averse miners prefer joining a pool over solo mining. The threshold $\gamma_{12}(f_1, f_2)$ captures the trade-off between competing pools and reflects their relative risk profiles through the term $D_{12}(f_1, f_2)$. When $D_{12} > 0$, pool 1 is more risky than pool 2 and is therefore preferred only by miners with relatively low risk aversion. Conversely, when $D_{12} < 0$, pool 1 offers lower variance and attracts the most risk-averse miners.*

This implies a monotone sorting of miners according to their risk preferences: the mining ecosystem endogenously segments into groups that differ only through their degree of risk aversion. Consequently, the allocation of miners across pools is fully determined by the relative positions of the thresholds γ_{10}, γ_{20}, and γ_{12}. The mechanism is illustrated in Figure 1, where the intersections of the score functions determine the relevant thresholds and the induced partition of the type space.

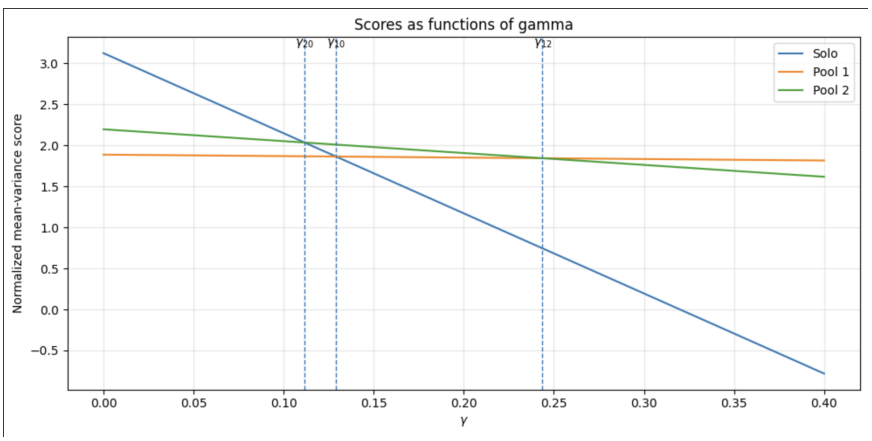

Figure 1: Mean–variance scores as functions of the risk-aversion parameter γ. The intersections of the score functions determine the thresholds γ_{10}, γ_{20}, and γ_{12}, which partition the support $\gamma \in [0, \bar{\gamma}]$ into regions where miners optimally choose solo mining, pool 1, or pool 2.

3.3.2 Expected participation and hashrate allocation

The threshold structure established in Proposition 1 allows for a tractable characterization of miner participation. Since miners are heterogeneous only through their risk-aversion parameter γ, and since the optimal choice is determined by threshold comparisons, the allocation of miners across alternatives can be obtained by measuring the corresponding intervals in the support $[0, \bar{\gamma}]$.
We denote by $p_k(f_1, f_2)$ the probability that a randomly selected miner joins option $k \in \{0, 1, 2\}$, where $k = 0$ corresponds to solo mining. Under the assumption that $\gamma \sim \text{Unif}[0, \bar{\gamma}]$, these probabilities are proportional to the length of the intervals identified in Proposition 1.

Proposition 2 (Participation probabilities)**.** *Let $(f_1, f_2) \in [0, 1)^2$ be a given fee profile. The participation probabilities are determined by the relative position of the thresholds $\gamma_{10}(f_1)$, $\gamma_{20}(f_2)$, and $\gamma_{12}(f_1, f_2)$.*

(i) Case $D_{12}(f_1, f_2) > 0$. *Pool 1 is preferred only by sufficiently weakly risk-averse miners. The participation probability of pool 1 is given by*

$$p_1(f_1, f_2) = \frac{\left[\min\{\bar{\gamma}, \gamma_{12}(f_1, f_2)\} - \max\{0, \gamma_{10}(f_1)\}\right]_+}{\bar{\gamma}}. \tag{24}$$

(ii) Case $D_{12}(f_1, f_2) < 0$. *Pool 1 is preferred by sufficiently risk-averse miners. The participation probability of pool 1 is given by*

$$p_1(f_1, f_2) = \frac{\left[\bar{\gamma} - \max\{0, \gamma_{10}(f_1), \gamma_{12}(f_1, f_2)\}\right]_+}{\bar{\gamma}}. \tag{25}$$

(iii) Case $D_{12}(f_1, f_2) = 0$. *The comparison between the two pools is independent of* γ. *If* $f_1 \leq f_2$, *pool 1 weakly dominates pool 2, and its participation probability reduces to*

$$p_1(f_1, f_2) = \frac{\left[\bar{\gamma} - \max\{0, \gamma_{10}(f_1)\}\right]_+}{\bar{\gamma}}. \tag{26}$$

Otherwise, $p_1(f_1, f_2) = 0$.

The participation probability of pool 2 is obtained symmetrically, and the share of solo miners satisfies

$$p_0(f_1, f_2) = 1 - p_1(f_1, f_2) - p_2(f_1, f_2). \tag{27}$$

Proof. By Proposition 1, each option $k \in \{0, 1, 2\}$ is chosen on an interval $I_k(f_1, f_2) \subseteq [0, \bar{\gamma}]$ defined by the corresponding threshold conditions. Since $\gamma \sim \mathrm{Unif}[0, \bar{\gamma}]$, we have

$$p_k(f_1, f_2) = \mathbb{P}(\gamma \in I(f_1, f_2)) = \frac{\mathrm{Leb}(I(f_1, f_2))}{\bar{\gamma}},$$

where $\mathrm{Leb}(\cdot)$ denotes the Lebesgue measure. The expressions follow by identifying the relevant intervals in each case according to the ordering of the thresholds γ_{10}, γ_{20}, and γ_{12}. □

Let n denote the total number of miners. The expected number of miners joining option k is given by

$$\mathbb{E}[N_k(f_1, f_2)] = n\, p_k(f_1, f_2), \qquad k \in \{0, 1, 2\}. \tag{28}$$

To relate participation to aggregate mining power, we assume that individual hashrates $(\lambda_i)_{i=1}^n$ are independent of risk preferences and satisfy

$$\sum_{i=1}^{n} \lambda_i = \lambda, \qquad \lambda_i \perp \gamma_i. \tag{29}$$

Under this assumption, expected hashrates are proportional to participation probabilities:

$$\mathbb{E}[H_k(f_1, f_2)] = \lambda\, p_k(f_1, f_2), \qquad k \in \{0, 1, 2\}. \tag{30}$$

This characterization provides a complete description of the demand side of the mining market. The participation probabilities are piecewise-defined functions of the fee profile (f_1, f_2), with kinks

arising when one of the thresholds γ_{10}, γ_{20}, or γ_{12} reaches the boundary of the support $[0, \bar{\gamma}]$. As a consequence, marginal changes in fees may induce discrete shifts in participation through changes in the ordering of these thresholds. Moreover, the aggregation step shows that the distribution of hashrate across pools is entirely summarized by the probabilities (p_0, p_1, p_2) and the total available hashrate λ. This reduction will be instrumental in the subsequent analysis of pool managers' optimal fee-setting behavior.

3.3.3 Expected payoffs of pool managers

Using the aggregation result $\mathbb{E}[H_k(f_1, f_2)] = \lambda\, p_k(f_1, f_2)$ established in the previous subsection, the expected payoff of pool manager $k \in \{1, 2\}$ can be expressed directly as a function of the fee profile (f_1, f_2). Substituting into (16) yields

$$\mathbb{E}[J_k(f_1, f_2)] = y_k + \lambda\, p_k(f_1, f_2) \frac{b}{1 + \delta_k} B_k(f_k), \tag{31}$$

where

$$B_k(f_k) = f_k(1 + \delta_k) - \delta_k - \eta_k b\Big(\delta_k(1 - f_k)^2 + (1 - (1 - f_k)\delta_k)^2\Big). \tag{32}$$

The representation (31) makes explicit that the manager's payoff is linear in the participation probability $p_k(f_1, f_2)$ and therefore in the share of total hashrate attracted by the pool. The term $B_k(f_k)$ captures the contribution of the fee choice to the payoff per unit of hashrate, incorporating both the expected revenue and the variance penalty induced by the mean–variance criterion.

This formulation highlights the trade-off faced by pool managers. Increasing the fee f_k raises the expected revenue component of $B_k(f_k)$, but simultaneously reduces the participation probability $p_k(f_1, f_2)$ by making the pool less attractive to miners. In addition, the quadratic terms in $B_k(f_k)$ reflect the impact of risk preferences, penalizing fee levels that increase the variability of the surplus process.

Consequently, the competition between pools can be interpreted as a pricing game with endogenous demand: each manager chooses a fee to balance the margin extracted from participating miners against the total hashrate attracted. This reduced-form representation provides the basis for the characterization of the Nash equilibrium in fees developed in the next subsection.

3.3.4 Nash equilibrium in a fee competition

We now formalize the strategic interaction between pool managers. The competitive setting considered in this paper is a static two-stage game. In the first stage, each pool manager $k \in \{1, 2\}$ simultaneously chooses a fee $f_k \in [0, 1)$. In the second stage, miners observe the fee profile (f_1, f_2) and select the option that maximizes their mean–variance criterion. This induces participation probabilities $p_k(f_1, f_2)$ and determines the allocation of hashrate across pools.

The resulting interaction can therefore be interpreted as a pricing game with endogenous demand, in which each manager anticipates the miners' response when selecting fees.

Definition 1 (Nash equilibrium)**.** *A fee profile $(f_1^*, f_2^*) \in [0,1)^2$ is a Nash equilibrium if, for each $k \in \{1,2\}$, the fee f_k^* solves*

$$\sup_{f_k \in [0,1)} \mathbb{E}[J_k(f_k, f_{-k}^*)], \tag{33}$$

where f_{-k}^* denotes the equilibrium fee of the competing pool. Equivalently, a Nash equilibrium is a fee pair such that no pool manager can increase its expected payoff by unilaterally deviating from f_k^*, given the fee chosen by the other pool.

The main difficulty in characterizing such an equilibrium lies in the fact that the participation probabilities $p_k(f_1, f_2)$ are piecewise-defined functions of the fee profile. Their analytical expressions depend on the ordering of the thresholds γ_{10}, γ_{20}, and γ_{12}, as well as on the sign of $D_{12}(f_1, f_2)$. As a consequence, the expected payoff functions are generally not globally smooth, and the analysis must be conducted region by region.

We focus on an interior coexistence region in which both pools attract a strictly positive fraction of miners. Consider the region defined by

$$D_{12}(f_1, f_2) < 0, \qquad 0 < \gamma_{12}(f_1, f_2) < \bar{\gamma}, \qquad \gamma_{12}(f_1, f_2) \geq \gamma_{10}(f_1), \qquad \gamma_{12}(f_1, f_2) \geq \gamma_{20}(f_2). \tag{34}$$

In this configuration, the participation probabilities simplify to

$$p_1(f_1, f_2) = \frac{\bar{\gamma} - \gamma_{12}(f_1, f_2)}{\bar{\gamma}}, \qquad p_2(f_1, f_2) = \frac{\gamma_{12}(f_1, f_2) - \gamma_{20}(f_2)}{\bar{\gamma}}. \tag{35}$$

Within this region, the expected payoff functions are differentiable, and interior equilibria can be characterized using first-order conditions. For manager k, the optimality condition is given by

$$\frac{\partial \mathbb{E}[J_k]}{\partial f_k}(f_1, f_2) = \lambda \frac{b}{1+\delta_k} \left[\frac{\partial p_k}{\partial f_k}(f_1, f_2) B_k(f_k) + p_k(f_1, f_2) B_k'(f_k) \right] = 0. \tag{36}$$

This condition admits a natural economic interpretation. The first term captures the marginal effect of the fee on participation (demand effect), while the second term captures the marginal effect on per-unit profitability (margin effect). At equilibrium, these two forces exactly offset each other. Using the expressions of $p_k(f_1, f_2)$ in the coexistence region, one obtains explicit formulas for the derivatives $\partial p_k / \partial f_k$ and $B_k'(f_k)$ (see Appendix A), and hence a system of two coupled equations

$$\begin{aligned} \frac{\partial \mathbb{E}[J_1]}{\partial f_1}(f_1^*, f_2^*) &= 0, \\ \frac{\partial \mathbb{E}[J_2]}{\partial f_2}(f_1^*, f_2^*) &= 0. \end{aligned} \tag{37}$$

A candidate equilibrium (f_1^*, f_2^*) must therefore solve this system. However, this is not sufficient. Since the expressions for p_1 and p_2 are valid only within the region defined by (34), any solution

must also satisfy the consistency conditions

$$D_{12}(f_1^*, f_2^*) < 0, \qquad 0 < \gamma_{12}(f_1^*, f_2^*) < \bar{\gamma}, \qquad \gamma_{12}(f_1^*, f_2^*) \geq \gamma_{10}(f_1^*), \qquad \gamma_{12}(f_1^*, f_2^*) \geq \gamma_{20}(f_2^*). \tag{38}$$

These constraints ensure that the candidate solution is consistent with the underlying participation structure. If any of these conditions fails, the equilibrium does not belong to the interior coexistence region, and the analysis must be carried out in the corresponding alternative region of the fee space.

Example 1. *We consider a fee-setting game with two PPS pools and a continuum of miners whose risk-aversion parameter is uniformly distributed on* $[0, \bar{\gamma}]$. *The parameters are set as*

$$b = 3.125, \qquad \bar{\gamma} = 0.4, \qquad \lambda = 100,$$

$$\delta_1 = 0.008, \qquad \delta_2 = 0.08, \qquad \eta_1 = 0.01, \qquad \eta_2 = 0.001, \qquad y_1 = y_2 = 100.$$

Solving the first-order conditions in the interior coexistence region yields the candidate fee profile

$$f_1^* = 9.9949901714\%, \qquad f_2^* = 8.5458873160\%,$$

and the solution is consistent with the interior coexistence region. The corresponding allocation of hashrate across solo mining and the two pools is illustrated in Figure 2, which reports the proportion of total hashrate collected by each option.

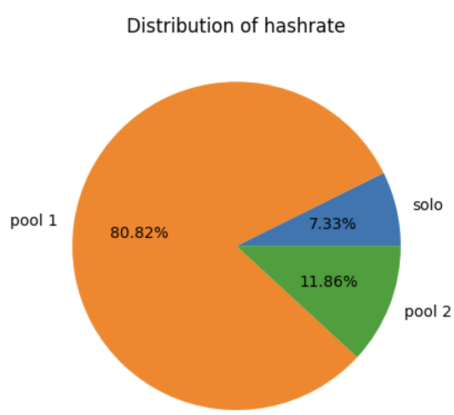

Figure 2: Distribution of the hashrate power among solo, pool 1 and pool2.

The equilibrium in Example 1 therefore reflects the endogenous segmentation of miners according to their risk preferences. This example also highlights a limitation of the mean–variance framework in the present setting. Since each miner selects the single alternative that maximizes its objective function, the allocation rule is of all-or-nothing type: a miner can join one pool or mine solo, but cannot diversify its hashrate across several pools. From an economic perspective, this restriction is strong, as diversification is a natural response to risk. This limitation motivates the introduction of the dividend-based approach in the next chapter, which provides a richer framework for the analysis of mining decisions and pool competition.

Acknowledgmements

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A Derivatives of Participation Functions and Payoff Components

A.1 Interior coexistence region

We consider the interior coexistence region characterized by

$$D_{12}(f_1, f_2) < 0, \qquad 0 < \gamma_{12}(f_1, f_2) < \bar{\gamma}, \qquad \gamma_{12}(f_1, f_2) \geq \gamma_{10}(f_1), \qquad \gamma_{20}(f_2) \leq \gamma_{12}(f_1, f_2).$$

In this region, participation probabilities are given by

$$p_1(f_1, f_2) = \frac{\bar{\gamma} - \gamma_{12}(f_1, f_2)}{\bar{\gamma}}, \qquad p_2(f_1, f_2) = \frac{\gamma_{12}(f_1, f_2) - \gamma_{20}(f_2)}{\bar{\gamma}}. \tag{39}$$

A.2 Derivative of γ_{12}

Recall that

$$\gamma_{12}(f_1, f_2) = \frac{f_2 - f_1}{bD_{12}(f_1, f_2)}, \qquad D_{12}(f_1, f_2) = \delta_1(1 - f_1)^2 - \delta_2(1 - f_2)^2. \tag{40}$$

Differentiating with respect to f_1 yields

$$\frac{\partial \gamma_{12}}{\partial f_1} = \frac{-bD_{12}(f_1, f_2) + 2\delta_1(1 - f_1)b(f_2 - f_1)}{b^2 D_{12}(f_1, f_2)^2}. \tag{41}$$

Differentiating with respect to f_2 yields

$$\frac{\partial \gamma_{12}}{\partial f_2} = \frac{bD_{12}(f_1, f_2) - 2\delta_2(1 - f_2)b(f_2 - f_1)}{b^2 D_{12}(f_1, f_2)^2}. \tag{42}$$

A.3 Derivative of γ_{20}

Recall that

$$\gamma_{20}(f_2) = \frac{f_2}{b\,(1 - \delta_2(1 - f_2)^2)}. \tag{43}$$

Differentiating with respect to f_2 yields

$$\frac{\partial \gamma_{20}}{\partial f_2} = \frac{1 - \delta_2(1 - f_2)^2 - 2\delta_2 f_2(1 - f_2)}{b\,(1 - \delta_2(1 - f_2)^2)^2}. \tag{44}$$

A.4 Derivatives of participation probabilities

Using the previous results, we obtain

$$\frac{\partial p_1}{\partial f_1} = -\frac{1}{\bar{\gamma}} \frac{\partial \gamma_{12}}{\partial f_1} = \frac{D_{12}(f_1, f_2) - 2\delta_1(1 - f_1)(f_2 - f_1)}{b\bar{\gamma}\, D_{12}(f_1, f_2)^2}. \tag{45}$$

Similarly,

$$\frac{\partial p_2}{\partial f_2} = \frac{1}{\bar{\gamma}} \left(\frac{\partial \gamma_{12}}{\partial f_2} - \frac{\partial \gamma_{20}}{\partial f_2} \right), \tag{46}$$

that is,

$$\frac{\partial p_2}{\partial f_2} = \frac{1}{\bar{\gamma}} \left[\frac{D_{12}(f_1, f_2) - 2\delta_2(1 - f_2)(f_2 - f_1)}{bD_{12}(f_1, f_2)^2} - \frac{1 - \delta_2(1 - f_2)^2 - 2\delta_2 f_2(1 - f_2)}{b\,(1 - \delta_2(1 - f_2)^2)^2} \right]. \tag{47}$$

A.5 Derivatives of $B_k(f_k)$

Recall that

$$B_k(f_k) = f_k(1 + \delta_k) - \delta_k - \eta_k b \Big(\delta_k(1 - f_k)^2 + (1 - (1 - f_k)\delta_k)^2 \Big). \tag{48}$$

Differentiating with respect to f_k yields

$$B_k'(f_k) = (1 + \delta_k) - 2\eta_k b \delta_k \Big((1 + \delta_k) f_k - \delta_k \Big). \tag{49}$$